

Facility: SALEM 1 & 2 Scenario No.: ESG-1 Op-Test No.: 11-01 NRC

Examiners: _____ Operators: _____

Initial Conditions: 100% power, EOL. PZR Pressure channel II is O/S for calibration.

Turnover: Maintain current power. Perform IST Room Cooler Valves Modes 1-6 IAW S2.OP-ST.SW-0014.

Event No.	Malf. No.	Event Type*	Event Description
1		N CRS/PO	Safety Related Room cooler surveillance (TS)
2	SG0095A	I	SG NR Ch failure (TS)
3	TA0314 RD0061	R, C ALL	Stator Water runback w/ subsequent auto rod control failure
4	O/R CF12	C CRS/PO	MS10 failure in automatic
5	RP0176A C812/C809 RC0001	M ALL	Inadvertent SI, 2C 4KV vital bus UV following Rx trip, LBLOCA
6	RP318A1,2	C ALL	Lo Head ECCS fails to actuate
7	RP0277A,B	C CRS RO	CS fails to actuate
8	CV62B	C RO	Charging pump cavitation
			CT's: Init Cont Spray, initiate Lo head ECCS, xfer to CLR, trip cavitating charging pp.

* (N)ormal, (R)eactivity, (I)nstrument, (C)omponent, (M)ajor

SIMULATOR EXAMINATION SCENARIO GUIDE

SCENARIO TITLE: 11-01 NRC ESG-1

SCENARIO NUMBER: 11-01 NRC ESG-1

EFFECTIVE DATE: See Approval Dates below

EXPECTED DURATION: 70 minutes

REVISION NUMBER: 01

PROGRAM: ☐ L.O. REQUAL
☒ INITIAL LICENSE
☐ STA
☐ OTHER _____

Revision Summary

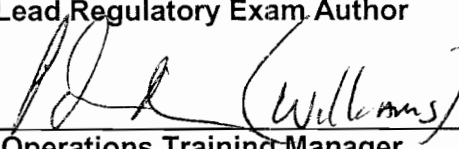
New issue for 09-01 NRC Exam (Spare Scenario not used)

Rev. 01 – Modified to SBT format.

PREPARED BY: G Gauding
Lead Regulatory Exam Author

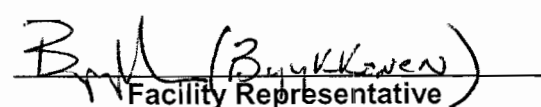
08-4-12

Date

APPROVED BY:  (Williams)
Operations Training Manager

9/12/12

Date

APPROVED BY:  (Bryukhov)
Facility Representative

9/6/12

Date

I. OBJECTIVES

- A. Given the unit with a condition requiring a stator water runback, the crew will take action to verify or perform the actions needed for the runback, IAW approved station procedures.
- B. Given a situation requiring rapid boration, the crew will initiate a rapid boration and terminate when required IAW approved station procedures.
- C. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with the approved station procedures.
- D. Given the order or indications of a reactor trip perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with the approved station procedures.
- E. Given indication of a safety injection DIRECT the response to the safety injection in accordance with the approved station procedures.
- F. Given the order or indications of a safety injection perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.
- G. Given a safety injection has occurred and equipment has failed to START equipment that has failed to automatically start in accordance with the approved station procedures.
- H. Given indication of a loss of coolant accident (LOCA) DIRECT the immediate response to the LOCA in accordance with the approved station procedures.
- I. Given the order or indications of a loss of coolant accident (LOCA), complete actions as the nuclear control operator to PERFORM the immediate response to the LOCA in accordance with the approved station procedures.
- J. Given a loss of coolant accident (LOCA) and plant conditions to support cold leg recirculation DIRECT actions to transfer to cold leg recirculation DIRECT actions to transfer to cold leg recirculation.
- K. Given the order or a loss of coolant accident (LOCA) and plant conditions to support cold leg recirculation perform actions as the nuclear control operator to TRANSFER to cold leg recirculation in accordance with the approved station procedures.
- L. Given indication of excessive containment pressure DIRECT the response to the excessive pressure in accordance with the approved station procedures.
- R. Given the order or indications of excessive containment pressure perform actions as the nuclear control operator to RESPOND to the excessive pressure in accordance with the approved station procedures.
- S. Given the unit in the EOPs and an excessive containment pressure the crew will reduce containment pressure IAW approved station procedures.

II. MAJOR EVENTS

- A. Room cooler surveillance unsat
- B. 21 SG Ch II NR level fails high.
- C. Automatic Stator Water Runback
- D. MS10 failure
- E. LBLOCA

III. SCENARIO SUMMARY

- A. The crew will take the turnover at 100% power, EOL. I&C is performing a channel calibration of PZR Pressure Channel II. 2PR7 is shut with power removed to comply with TSAS 3.4.5.b. All bistables associated with PZR Pressure Channel II are tripped IAW S2.IC.-CC.RCP-0018, 2PT-456 Pressurizer Pressure Protection Channel II. Orders for the shift are to complete IST of Room Cooler valves (RHR pump room coolers) IAW S2.OP-ST.SW-0014, Inservice Testing Room Cooler Valves Modes 1-6 for the RHR pump room coolers. Maintain 100% power.
- B. After assuming the watch, the crew will test the RHR pump room coolers. The 21SW153 room cooler SW valve stroke time will be in the Required Evaluation Range. The CRS will either declare the valve inoperable, or retest the valve as called for in the procedure. If a second test is performed, the stroke time will fall in the Required Action Range, and the CRS will declare the valve inoperable. IAW S2.OP-SO.SW-0005, Service Water System Operation, the RHR pump must be declared inoperable when its room cooler is inoperable. (page 104). The CRS enters TS 3.5.2 for only 1 train of ECCS.
- C. After completing the surveillance, 21 SG NR level Ch. II fails high. The crew will verify no ADFCS malfunction and the CRS will enter the appropriate Tech Spec.
- D. A Stator Water Main Turbine runback will initiate and the runback will terminate at ~900 Mwe when the initiating condition clears. Control Rod speed fails to minimum, and operators will place Rod control in manual to insert rods. Operators will ensure positive control of the Main Turbine and borate the RCS in conjunction with rod insertion to restore Tavg and AFD.
- E. Once the plant has been stabilized, 21MS10 fails ~ 33% open in auto. The PO will take manual control and shut 21MS10.
- F. A false Safety Injection signal is generated, and the Rx automatically trips. Operators will backup the Rx trip and Safety Injection actuation.
- G. 2C 4KV Vital bus will deenergize upon the Rx trip, and the SEC will load 2C vital bus on its EDG. 21 CCW pump will trip, and 23 CCW pump control will transfer and remain in manual. 22 charging pump will not start on SEC initiation, and operators will block the SEC and start 22 charging pump. A LBLOCA occurs 5 minutes after the Rx trips. Automatic Phase B and Containment Spray will not occur, and operators will manually initiate Cont Spray and manually start Containment Spray pumps. Operators will perform diagnostics in TRIP-1 and transition to LOCA-1. 21 charging pump will trip 15 minutes after the Rx trip.

- H. The crew will transition to LOCA-3 when RWST level reaches 15.2 ft. The crew will perform the transfer to CL Recirc with all 4KV vital busses energized and 2C bus energized from its EDG. During the transfer, 22 charging pump will cavitate when 2SJ1 and 2SJ2 are shut, and operators will stop the pump. If the pump is not stopped, it will fail, and a LOCA outside containment will occur.
- I. The scenario will terminate when 22 charging pump is tripped, or after indications of CVCS system piping failure have occurred if the pump was not secured.

IV. INITIAL CONDITIONS

_____ IC-3 or Pre-snapped IC-241

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

Initial	Description
_____ 1	RH1 and RH2 C/T
_____ 2	RH18s C/T
_____ 3	VC1and VC4 C/T
_____ 4	RCPs (SELF CHECK)
_____ 5	RTBs (SELF CHECK)
_____ 6	MS167s (SELF CHECK)
_____ 7	500 KV SWYD (SELF CHECK)
_____ 8	SGFP Trip (SELF CHECK)
_____ 9	23 CV PP (SELF CHECK)
_____ 10	OHA G-20 in alarm.
_____ 11	2PR7 shut and deenergized.
_____ 12	S2.IC-CC-RCP-0018, 2PT-456 Pressurizer Pressure Protection Channel II, Exhibit 1 signed and on NCO desk or provided to crew for turnover.
_____ 13	Marked up S2.OP-ST.SW-0014, Inservice Testing Room Cooler Valves Modes 1-6 and S2.RA-ST.SW-0014 Rev. 29, Acceptance Criteria, provided to crew with turnover.
_____ 14	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:			
Initial	ET #	Description	
	1	EVENT ACTION: COMMAND: PURPOSE:	MONP254 <10 // CONT ROD BANK C < 10 (RX TRIP) <update as needed>
	3	EVENT ACTION: COMMAND: PURPOSE:	gbc18axd < 0.851 // (2D5) CONTROL ROD POSITION DMF TA0314 <update as needed>
	5	EVENT ACTION: COMMAND: PURPOSE:	kb117lck // 2SJ2 RWST TO CHG PUMP CLOSE <update as needed>
	7	EVENT ACTION: COMMAND: PURPOSE:	MONP254 <10 // CONT ROD BANK C < 10 (RX TRIP) <update as needed>
	11	EVENT ACTION: COMMAND: PURPOSE:	gcf11bx1 > 0.2 // 21MS10 VALVE DEM DOR KCF12LW1 <update as needed>

MALFUNCTIONS:						
SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
01	AN0302 SER 302 FAILS - :E35 2PR2 PROTECTION CH II LOOP UNSAFE/IN TEST	N/A	N/A	N/A	N/A	SER POINT FAILS/OVRD TO ON
02	AN0328 SER 328 FAILS - :A12 REACTOR PROT SYS CHANNEL II INSTRUMENT LOOP IN TEST	N/A	N/A	N/A	N/A	SER POINT FAILS/OVRD TO ON
03	TA0314 STATOR WATER FLOW SIGNAL FAIL	N/A	N/A	N/A	RT-3	
04	SG0095A 21 SG LVL XMTR (519) CH II FAIL	N/A	N/A	N/A	RT-1	120
05	RC0001A RCS RUPTURE OF RC LOOP 21	00:05:00	N/A	N/A	ET-7	
06	RP318S2 22 CHRGR PMP FAILS TO STRT ON SEC	N/A	N/A	N/A	N/A	
07	CV0208A 21 CHARGING PUMP TRIP	00:15:00	N/A	N/A	ET-7	
08	RP0277A AUTO CNT SPRY FAILS TO ACT,TRN	N/A	N/A	N/A	N/A	
09	RP0277B AUTO CNT SPRY FAILS TO ACT,TRN	N/A	N/A	N/A	N/A	
10	CV0043 CHARGING LINE LEAK IN AUX BLDG	N/A	N/A	N/A	RT-9	550
11	CV0208B 22 CHARGING PUMP TRIP	00:02:00	N/A	N/A	RT-9	
12	AN0289 SER 289 FAILS - :H45 GENERATOR STATOR COOLING SYSTEM TROUBLE	N/A	N/A	N/A	RT-11	SER POINT FAILS/OVRD TO OFF
13	RM0210B1 AREA RAD MONITOR 2R4 FAILS ANYWHERE	00:00:30	N/A	N/A	RT-9	15
14	AN0737 SER 737 FAILS - :A6 RMS HI RAD OR TRBL-CHG PUMPROOM CH 2R4	00:00:30	N/A	N/A	RT-9	SER POINT FAILS/OVRD TO ON
15	CC0172A 21 COMPONENT COOLING PUMP TRIP	00:00:30	N/A	N/A	ET-1	
16	RP318A1 RHR PUMP 21 Fails to Start on SEC	N/A	N/A	N/A	N/A	
17	RP318A2 RHR PUMP 22 Fails to Start on SEC	N/A	N/A	N/A	N/A	
18	AN0269 SER 269 FAILS - :G20 SEAL & CYLINDER HEATING STEAM SYSTEM TROUBLE	N/A	N/A	N/A	N/A	SER POINT FAILS/OVRD TO ON
19	RD0061 ROD SPEED CONTROL PROGRAM FAILS	N/A	N/A	N/A	ET-3	8
20	RP0176A FALSE SAFETY INJ ACT SIGNAL A	N/A	N/A	N/A	RT-7	

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
01	PR02D PZR HI PR RX TRP CH 2 PC456A BS	N/A	N/A	N/A	N/A	TRIP
02	PR13D PZR LO PR BLK SI CH 2 PC456B BS	N/A	N/A	N/A	N/A	TRIP
03	PR06D PZR LO PR RX TRP CH 2 PC456C BS	N/A	N/A	N/A	N/A	TRIP
04	PR10D PZR LOW PRESS SI CH 2 PC456D BS	N/A	N/A	N/A	N/A	TRIP
05	RC02D OT DELTA T TRIP CH II BS (421C)	N/A	N/A	N/A	N/A	TRIP
06	RC06D OT DELTA T R/BCK CH II BS (421D)	N/A	N/A	N/A	N/A	TRIP
07	PR35D PORV STOP VALVE 2PR7 TAGGED	N/A	N/A	N/A	N/A	TAGGED
08	CV62B 22 CHG PUMP SUCTION VALVE 2CV49	N/A	N/A	N/A	ET-5	.000045

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/Severity
01	A604 B DI 23 CC PUMP - MANUAL	N/A	N/A	N/A	ET-7	ON
02	A604 A DI 23 CC PUMP - AUTO	N/A	N/A	N/A	ET-7	OFF
03	C812 F DI 24CSD 2C VITAL BUS FEEDER-OPEN	N/A	N/A	N/A	N/A	ON
04	C809 F DI 23CSD 2C VITAL BUS FEEDER-OPEN	N/A	N/A	N/A	ET-7	ON
05	A604 E DI 23 CC PUMP - START	N/A	N/A	N/A	ET-7	OFF
06	CF12 D DI 21MS10 PRESSURE SET PT. DECREASE	N/A	N/A	N/A	RT-5	ON

OTHER CONDITIONS:

Description

1. PZR Pressure Channel IV selected for Alarm

V. SEQUENCE OF EVENTS

- A. State shift job assignments.
- B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).
- C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".
- D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. Valve surveillance with Unsat result			
	CRS directs PO to commence testing of 21 RHR pump room cooler.		
	PO records as found position (auto) of room cooler on Attachment 2.		
	PO contacts field operators for performance of valve stroke time monitoring.		
	PO and field operators test 21SW153.		
	PO reports stroke time is in the Required Evaluation Range.		
	CRS either directs valve to be retested, or declares valve inoperable.		
Role Play: Report after the room cooler is started that the 21SW153 stroke time was 5.2 seconds.			
Role Play: If PO retests 21SW153, report stroke time is 5.7 seconds			
	After the second stroke time test is performed, CRS declares the 2SW153 inoperable if not already done.		
	CRS declares 21 RHR pump inoperable, and enters TSAS 3.5.2 for less than 2 full ECCS trains.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: S2.OP-SO.SW-0001, Service Water System Operation, Attachment 2, SW System Operability Guidelines, page 104 discusses RHR Pump Room Cooler and RHR pump operability. Proceed to next event on direction from Lead Evaluator. 2. 21 SG NR level Channel II fails high			
Simulator Operator: Insert RT-1 on direction from Lead Evaluator MALF: SG0095A 21 SG lvl xmtr (519) CH II fails Final Value: 120			
	PO announces OHA G-15 ADFCS Trouble, and 21 SG Isolation Level Hi-Hi console alarms as unexpected.		
	PO refers to ARP.		
	PO identifies 21 SG NR level CH II has failed high.		
	PO verifies all ADFCS parameters are normal and still being controlled in automatic.		
	CRS contacts I&C or WCC and informs them of channel failure.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Proceed to next event after Tech Spec call at direction of Lead Evaluator. 3. Stator Water Runback	CRS enters TSAS 3.3.1.1 Action 6, and 3.3.2.1 action b 19.		
Simulator Operator: Insert RT-3 on direction from Lead Evaluator. MALF: TA0314 Stator Water Flow Signal Fail Final Value: True			
Simulator Operator: Ensure ET-3 is True when control bank D lowers to 192 steps. This deletes the Main Turbine runback and inserts the rod speed failure to 8 spm.	PO announces OHA's H-38 GEN STAT RUNBACK, and H-45 GEN STAT CLG SYS TRBL as unexpected.		
	PO reports the Main Turbine is running back.		
	RO monitors primary plant and ensures control rods start stepping inward in auto when demanded due to power mismatch and Tref-Tavg deviation OR inserts rods if in Manual.		
Note: The runback was initiated at			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
30% per minute, but the Main Turbine Governor valves respond slower than that. When the load reduction signal clears, the DEHC reference will stop changing, but the turbine governor valves will still be closing, and Mwe will lower another ~100 before reference equals actual.			
	RO reports control rod speed is not consistent with temperature and power deviation, and receives permission from CRS to place control rods in manual.		
	RO inserts control rods in manual to maintain Tavg-Tref mismatch in the band given by CRS.		
	PO starts standby Stator Cooling Water pump IAW ARP.		
	RO/PO initiates boration.		
	CRS dispatches an operator to the Stator Water Panel 968 to investigate and determine panel alarm status IAW ARP for OHA H-38.		
Role Play: 5 minutes after being dispatched, insert RT-11 which will clear OHA H-45. Then call control room as NEO and report: Panel 968 alarm A-1L, Inlet Flow Low was flashing, but cleared when I acknowledged it. Inlet flow is			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
reading 600 gpm. There are currently NO alarms on the panel.			
	PO verifies that when OHA H-38 clears, the DEHC reference stops lowering.		
	IF OHA E-16 ROD INSERT LMT LO-LO annunciates, THEN RO/PO initiates Emergency Boration IAW S2.OP-SO.CVC-0008 as directed by ARP by: - Determining that there is no set time duration for the boration per step 5.1.1. - Starts at least one Boric Acid Transfer pump in FAST speed. - Ensures charging flow is > 75 gpm. - Opens 2CV175 Rapid Borate Stop Valve - Closes 21 and 22CV160, Recirc Vlvs - Ensures boric acid flow is >36 gpm on 2FI-113A. OR - Ensures charging flow is >75 gpm on 2FI-128B - Starts at least one Boric Acid Transfer pump in FAST speed. - Ensures the 2CV179 and 2CV181 shut. - Manually open 2CV172. - Open 2CV185. - Close 2CV160 and 2CV161. - Ensure Boric Acid flow >36gpm on 2FI-110A.		

Note: Emergency Boration is to continue until control rods are at least 2 steps higher than the RIL,

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
and OHA E-16 clears.			
	PO ensures generator MVARs within band.		
	CRS enters S2.OP-AB.LOAD-0001, Rapid Load Reduction.		
	IF emergency boration was initiated, THEN WHEN OHA E-16 clears, RO stops the emergency boration by: - Stopping the Boric Acid Transfer pump started above - Closing the 2CV175 - Throttling open 21 and 22CV160 as required. OR - Place the running Boric Acid pump in Slow Speed. - Close the 2CV172 and 2CV185. - Return the CVCS Makeup system to AUTO. - Throttle open 2CV160 and 2CV161.		
	CRS enters TSAS 3.2.1 Action a.2 for AFD.		
Proceed to next event on direction from Lead Evaluator.			
4. MS10 failure			
Simulator Operator: Insert RT-5 at direction of Lead Evaluator.			
OVDI: CF12 21MS10 Pressure Setpoint Decrease PB ON.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Ensure ET-11 is true when 21MS10 position is 20% open. This deletes the OVDI and results in 21MS10 remaining ~40% open.			
Proceed to next event once 21MS10 is shut OR if crew trips the Rx. 5. LBLOCA	PO reports 21MS10 is opening with steam pressure setpoint lowering		
	PO either places 21MS10 in manual and shuts valve or raises steam pressure setpoint so that valve responds and shuts.		
	PO shuts 21MS10.		
	CRS enters S2.OP-AB.STM-0001, Excessive Steam Flow.		
Simulator Operator: Insert RT-7 on direction from Lead Evaluator. MALF: RP0176A False Safety Injection Signal Train A.			
	RO reports the Rx has tripped.		
Simulator Operator: Ensure ET-7 is TRUE upon the Rx trip. This will:			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Initiate the LBLOCA after 5 min. delay. Swap 23 CCW pump to manual to prevent it from auto starting. Trip 21 CCW pump after 30 sec delay. Open both infeed bkrs to 2C 4KV vital bus and cause EDG to load. Trip 21 charging pump after a 15 minute delay.			
	RO performs TRIP-1 immediate actions from memory: - Verifies Rx tripped - Verifies Rx trip confirmed - Trips the Main Turbine - Verifies at least one 4KV vital bus energized. - Reports Safety Injection actuated and backs up SI.		
	CRS reads immediate action steps to RO and RO verifies performance.		
	RO/PO reports 2C 4KV vital bus is energized from 2C EDG.		
	RO reports 22 charging pump, 21 and 22 RHR pumps failed to start.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
6. Lo Head ECCS fails to actuate	Crew blocks ALL SECs on 2RP1.		
	Crew resets Emergency loading on ALL SECs.		
	RO starts 22 Charging pump, 21 RHR pump, and 22 RHR pump.		
CT #1 (E-0—H) Manually start at least one RHR pump prior to transition out of EOP-TRIP-1.			
SAT UNSAT			
7. Containment Spray fails to actuate.	RO makes page announcement "Unit 2 Rx trip, Safety Injection."		
	PO reports all AFW pumps running, and requests permission to throttle AFW flow while maintaining 22E4 lbm/hr or 9% level in at least one SG NR level.		
	CRS directs PO to throttle AFW flow.		
	PO lowers 23 AFW pump speed to minimum, and throttles 21-24AF21's to maintain AFW flow ~ 22E4 lbm/hr.		
	RO/PO identifies that a LBLOCA has occurred by control board indications.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports control console indication of Containment Spray and Phase B actuation, but no Containment Spray system components have actuated.		
	CRS refers/returns to Step 11 and directs RO initiate Containment Spray by: - Initiate phase B and Spray actuation. - RO inserts BOTH keys into CS actuation and turns at same time. - RO reports spray valve realignment. - Start 21 and 22 CS pumps - RO inserts key and turns for 21 and 22 CS pumps and reports both CS pumps are running. - Initiate MSLI - RO reports MSLI initiated. - Stop 21-24 RCPs - RO depresses stop PB for all RCPs. - Ensure valve groups in Table D in safeguards position. - PO reports all Table D valves in safeguards position.		
CT #2 (E-0--E) Initiate Containment Spray prior to exiting TRIP-1.			
SAT _____ UNSAT _____			
	RO reports MSLI has been initiated.		
	CRS directs SM implement the ECG.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: The PO should NOT attempt to start 23 CCW pump here, since 2C vital 4kv bus is powered from the EDG and there is no direction in TRIP-1 to start the second CCW pump. IF the PO attempts to start 23 CCW pump, it will neither start nor backflash due to scenario setup override.	PO reports all 3 4KV vital busses are energized with 2C 4KV vital bus energized from 2C EDG.		
	RO reports control room ventilation is in Accident Pressurized mode.		
	RO reports 2 swgr supply and 1 swgr exhaust fans are running.		
	PO reports only 22 CCW pump is running.		
	CRS dispatches an operator to shut 2CC37 and 2CC48.		
	CREW performs ECCS flow evaluation and determines all ECCS pumps are injecting to RCS as expected for LBLOCA and cont press.		
	PO reports AFW status and is maintaining 22E4 lbm/hr or 15% (adverse cont numbers in effect) level in at least one SG NR level, then maintaining 15-33%.		
	RO reports all RCPs are stopped and MSLI		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: All SG pressures will be dropping due to the LBLOCA with 21 SG pressure lower than the remaining 3 SGs due to the RCS rupture being on that loop. Note: STA will report to the control room 10 minutes after being paged and begin monitoring CFSTs. Note: 21 charging pump will trip 15 minutes after the Rx trip. Note: Upon the transition out of TRIP-1, CFST monitoring becomes	actuated.		
	RO reports both RTBs open.		
	RO reports both PORVs shut with 2PR6 open and 2PR7 shut with power removed.		
	RO reports all RCPs are stopped.		
	RO ensures seal injection flow is maintained to all RCPs.		
	PO reports there are no faulted or ruptured SGs.		
	CRS transitions to LOCA-1 on containment radiation levels.		
	RO reports 21 charging pump has tripped.		

active.

Note: Depending on scenario timing, the FRCE Purple path may have cleared, and will clear within ~15 minutes of LBLOCA initiation.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports no faulted SG's.		
	PO reports AFW status and is maintaining 22E4 lbm/hr or 15% (adverse cont numbers in effect) level in at least one SG NR level, then maintaining 15-33%.		
	PO reports no indications of SGTR.		
	RO resets SI, Phase A, and Phase B.		
	RO opens both CA330s.		
	PO resets all SECs.		
	RO resets SGBD Sample Isolation Bypass and opens 21-24SS94s.		
	CRS directs chemistry to sample SGs for boron and activity.		
	RO reports both PORVs shut with 2PR6 open and 2PR7 shut with power removed.		
	RO reports subcooling is less than zero.		
	RO reports both CS pumps are running.		
	RO reports subcooling is less than zero.		
	RO reports RHR is not aligned for CL recirc.		
	RO reports RHR flow is > 300 gpm on both		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
If the crew arrives at Step 18.2 prior to RWST level of 15.2', they will enter a "do loop" with return to step 16. When RWST lo level alarm actuates, continue below.	SJ49s.		
	CRS requests assistance from TSC to restore off-site power to 2C 4KV vital bus.		
	RO reports both RHR pumps and both SJ44s are available.		
	RO/PO reports no rad monitors in Table C are in warning or alarm.		
	CRS directs chemistry to sample the RCS and analyze for failed fuel and boron.		
	CRS consults TSC about equipment availability to assist in long term recovery.		
	CRS verifies operators were dispatched in TRIP-1 to shut 2CC37 and 2CC48.		
	RO reports both RHR pumps are running.		
	RO reports RHR flow is > 300 gpm on both SJ49s.		
	RO reports RWST level.		
	RO reports RWST lo level alarm as expected alarm.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Log time of RWST LO-LO level alarm for CT#2 evaluation. ____:____:____			
	CRS transitions to LOCA-3.		
Simulator Operator: Upon entry into LOCA-3, DELETE I/O A604 OVDI for the START PB for 23 CCW pump and ENSURE it does NOT start. Be prepared to remotely STOP 23 CCW pump if required, as it affects scenario outcome and must NOT be running but capable of starting when crew arrives at step 28.2			
	RO reports both Cont Sump Level >62% are lit.		
	RO depresses Sump Auto Armed PBs for 21 and 22SJ44.		
	PO removes lockouts for 2SJ67, 2SJ68 and 2SJ69.		
	RO reports 21 and 22SJ44 are open.		
	RO reports both RHR pumps are running and shuts 2SJ69.		
	LOG TIME 2SJ69 Close is initiated: ____:____:____		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	LOG TIME Containment Spray pump stopped: ____:____:____		
CT# 3: (ES-1.3—A) Transfer to Cold Leg Recirculation such that at least one train of ECCS is in operation in the recirculation mode within the following time frames. 1. From RWST lo level alarm to initiating closed on 2SJ69 - ≤ 3.7 minutes 2. From RWST lo level alarm to one containment spray pump stop ≤ 5.5 minutes 3. From RWST lo level alarm to switchover completion (includes restarting ECCS pumps if stopped on RWST lo-lo level) - ≤ 11.2 minutes. SAT _____ UNSAT _____			
	RO reports SI is reset, and all SECs are reset.		
	PO reports 230V load centers are reset.		
	RO reports 2SJ69 is closed.		
	RO reports both CS pumps are operating and stops 22 CS pump.		
	RO shuts 21 and 22RH19s, and stops 23		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	charging pump if running.		
	CRS selects step 11 as the correct flowpath transition.		
	PO reports C bus is energized from its EDG.		
	RO reports 3 SW pumps are running.		
	RO reports both CCHXs are in service.		
	RO reports 23 CCW pump is NOT running.		
	RO reports 22 CS pump is NOT running.		
	RO starts 23 CCW pump.		
	RO reports 21 and 22CC16s are open.		
	RO shuts 2SJ67 and 2SJ68.		
	RO reports RH1 and RH2 are shut.		
	RO reports both RHR pumps are running and opens 21 and 22SJ45s.		
	RO reports 21 and 22SJ113s are open.		
	RO reports 22 charging pump and 21 and 22 SI pumps are running.		
LOG TIME ECCS Pumps verified running: ____:____:____			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
8. Charging pump caviatation	PO removes lockout from 2SJ30.		
	RO shuts 2SJ30, 2SJ1, and 2SJ2.		
Simulator Operator: Ensure ET-5 is TRUE when the close PB for 2SJ2 is depressed. This inserts the closure of 22 charging pump suction isolation valve 22CV49 and will cause 22 charging pump to cavitate. During validation, oscillations between 48-90 Amps were observed, and the pump did NOT trip on OC.			
	RO/PO recognizes 22 charging pump is cavitating by amperage indication, discharge flow and BIT flow, and discharge pressure oscillations.		
	RO trips 22 charging pump.		
CT#4 Trip the cavitating charging pump prior to pump/system piping damage. SAT UNSAT			
Simulator Operator: IF the crew does NOT trip 22 charging pump within 5 minutes of the initiation of cavitation, THEN insert RT-9 . This			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
simulates a failure of the charging system piping in the Aux Building resulting in a LOCA outside containment, then trips 22 charging pump after an additional 2 minute delay. MALF: CV0043 Charging Line Leak in Aux Bldg Final Value: 550 MALF: CV0208B 22 Chg Pump trip Delay: 2 minutes Final Value: True MALF: RM0210B1 Area Rad Monitor 2R4 fails anywhere Delay: 30 sec Final Value: 15 MALF: AN0737 SER 737 Fails - A6 RMS Hi Rad or Trouble			
TERMINATE the scenario when 22 charging pump has been secured, whether by operator action or when it trips 7 minutes after cavitating starts.			
	RO places 21 and 22RH29 controllers in Manual and ensures valves are shut.		
Note: Some of the following steps may be performed depending on when, or if, the crew recognizes the 22 charging pump cavitation, and should be evaluated accordingly.			
	CRS implements FRPs as necessary.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	CRS dispatches operator to tag SJ44 breakers open.		
	RO verifies Phase A and Phase B are reset, and CA330s are open.		
	PO stops 2A and 2B EDGs.		

VI. SCENARIO REFERENCES

- A. Alarm Response Procedures (Various)
- B. Technical Specifications
- C. Emergency Plan (ECG)
- D. OP-AA-101-111-1003, Use of Procedures
- E. 2-EOP-TRIP-1, Rx Trip or Safety Injection
- F. 2-EOP-LOCA-1 Loss of Coolant Accident
- G. 2-EOP-LOCA-3 Transfer to Cold Leg Recirculation
- H. S2.OP-AB.LOAD-0001 Rapid Load Reduction
- I. S2.OP-AB.RC-0001 Reactor Coolant System Leak
- J. S2.OP-ST.SW-0014, IST Room Cooler Valves Modes 1-6
- K. S2.RA-ST.SW-0014, IST Room Cooler Valves Modes 1-6 Acceptance Criteria
- L. S2.OP-SO.SW-0005, Service Water System Operation

ATTACHMENT 1 UNIT TWO PLANT STATUS TODAY

MODE: 1 POWER: 100 RCS BORON: 100 MWe 1200

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Core Burnup: 12,600 EFPD

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

3.4.5 action b for PZR Pressure Channel II calibration. 70 hours left in action time.

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

S2.IC-CC-RCP-0018, 2PT-456 Pressurizer Pressure Protection Channel II.

S2.OP-ST.SW-0014, Inservice Testing Room Cooler Valves Modes 1-6. 21 RHR pump room cooler is to be tested upon assuming the watch. All other room coolers have been tested SAT. 2 NEOs are stationed at 21 RHR pump room to allow coordination of testing via the plant page system.

ABNORMAL PLANT CONFIGURATIONS:

2PR7 shut and power removed to comply with TSAS 3.4.5.b for PZR Pressure Channel II being out of service for channel calibration.

PZR Pressure Channel IV selected for Alarm.

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.

No penalty minutes in the last 24 hrs.

PRIMARY:

3.3.1.1 action 6 and 3.3.2.1 action 19 for PZR Pressure Channel II PT-456.

SECONDARY:

OHA G-20 SEAL CVL HTG STM SYS TRBL is in alarm due to Gland Sealing Steam high pressure. NEO is investigating.

Heating steam is aligned to Unit 1.

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

None

ATTACHMENT 2**SIMULATOR READY FOR TRAINING CHECKLIST**

- ___ 1. Verify simulator is in "TRAIN" Load
- ___ 2. Simulator is in RUN
- ___ 3. Overhead Annunciator Horns ON
- ___ 4. All required computer terminals in operation
- ___ 5. Simulator clocks synchronized
- ___ 6. All tagged equipment properly secured and documented
- ___ 7. TSAS Status Board up-to-date
- ___ 8. Shift manning sheet available
- ___ 9. Procedures in progress open and signed-off to proper step
- ___ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced
- ___ 11. Required chart recorders advanced and ON (proper paper installed)
- ___ 12. All printers have adequate paper AND functional ribbon
- ___ 13. Required procedures clean
- ___ 14. Multiple color procedure pens available
- ___ 15. Required keys available
- ___ 16. Simulator cleared of unauthorized material/personnel
- ___ 17. All charts advanced to clean traces and chart recorders are on.
- ___ 18. Rod step counters correct (channel check) and reset as necessary
- ___ 19. Exam security set for simulator
- ___ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter
with Baseline Data filled out
- ___ 21. Shift logs available if required
- ___ 22. Recording Media available (if applicable)
- ___ 23. Ensure ECG classification is correct
- ___ 24. Reference verification performed with required documents available
- ___ 25. Verify phones disconnected from plant after drill.

ATTACHMENT 3**CRITICAL TASK METHODOLOGY**

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.

The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

- I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...
 - degradation of any barrier to fission product release
 - degraded emergency core cooling system (ECCS) or emergency power capacity
 - a violation of a safety limit
 - a violation of the facility license condition
 - incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
 - a significant reduction of safety margin beyond that irreparably introduced by the scenario
- II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...
 - effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
 - recognize a failure or an incorrect automatic actuation of an ESF system or component.
 - take one or more actions that would prevent a challenge to plant safety.
 - prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation).

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 11-01 NRC-ESG-1 REVIEWER: E Gallagher

Initials	Qualitative Attributes
EG	1. The scenario has clearly stated objectives in the scenario.
EG	2. The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events.
EG	3. The scenario consists mostly of related events.
EG	4. Each event description consists of: • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point
EG	5. No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event.
EG	6. The events are valid with regard to physics and thermodynamics.
EG	7. Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives.
EG	8. The simulator modeling is not altered.
EG	9. All crew competencies can be evaluated.
EG	10. The scenario has been validated.
EG	11. If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario.
EG	12. ESG-PSA Evaluation Form is completed for the scenario at the applicable facility.

ATTACHMENT 4**SIMULATOR SCENARIO REVIEW CHECKLIST**

Note: The quantitative attribute target ranges that are specified on the form are not absolute limitations; some scenarios may be an excellent evaluation tool, but may not fit within the ranges. A scenario that does not fit into these ranges shall be evaluated to ensure that the level of difficulty is appropriate. (ES-301 Section D.5.d)

Initial		Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)
GG	8	Total malfunctions inserted: 5-8
GG	4	Malfunctions that occur after EOP entry: 1-2
GG	2	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	2	EOPs entered/requiring substantive actions: 1-2
GG	1	EOP contingencies requiring substantive actions: 0-2
GG	4	Crew Critical Tasks: 2-3

COMMENTS:

ATTACHMENT 5

ESG CRITICAL TASKS

11-01 NRC ESG-1

CT#1 (E-0—H) Manually start at least one RHR pump prior to transition out of EOP-TRIP-1.

Basis- Failure to manually start at least one low-head ECCS pump under the postulated conditions constitutes "mis-operation or incorrect crew performance which leads to degraded...capacity." This is a critical task because of the LBLOCA which occurs, and lo head ECCS is required.

..

CT#2 (E-0--E) Manually actuate at least the minimum required complement of containment cooling equipment before exiting TRIP-1.

Basis- Failure to manually actuate the minimum required complement of containment cooling equipment under the postulated conditions constitutes a "demonstrated inability by the crew to recognize a failure/incorrect auto actuation of an ESF system or component." Failure to manually actuate containment spray represents a "demonstrated inability of the crew to effectively direct/manipulate ESF controls that would lead to a violation of the facility license condition."

- CT# 3: (ES-1.3—A) Transfer to Cold Leg Recirculation such that at least one train of ECCS is in operation in the recirculation mode within the following time frames:**
1. From RWST lo level alarm to initiating closed on 2SJ69 - ≤ 3.7 minutes.
 2. From RWST lo level alarm to one containment spray pump stop- ≤ 5.5 minutes.
 3. From RWST lo level alarm to switchover completion (includes restarting ECCS pumps if stopped on RWST lo-lo level) - ≤ 11.2 minutes.

Basis: Failure to establish cold leg recirculation within the time frame specified in the FSAR is a "demonstrated inability of the crew to effectively direct/manipulate ESF controls that would lead to a violation of the facility license condition."

CT#4: Trip the cavitating Charging Pump prior to pump/system piping damage.

Basis: Failure to trip a cavitating charging pump during the transfer to cold leg recirculation when indications of cavitation are present, leads to the possibility of system damage and the advent of a loss of coolant outside the containment building. (As is the case in this scenario.)

ATTACHMENT 6

ESG-PSA RELATIONSHIP EVALUATION

EVENTS LEADING TO CORE DAMAGE

<u>Y/N</u>	<u>Event</u>	<u>Y/N</u>	<u>Event</u>
<u>N</u>	TRANSIENTS with PCS Unavailable	<u>N</u>	Loss of Service Water
<u>N</u>	Steam Generator Tube Rupture	<u>N</u>	Loss of CCW
<u>N</u>	Loss of Offsite Power	<u>N</u>	Loss of Control Air
<u>N</u>	Loss of Switchgear and Pen Area Ventilation	<u>N</u>	Station Black Out
<u>Y</u>	LOCA		

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>	<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>
<u>N</u>	Containment Sump Strainers	<u>N</u>	Gas Turbine
<u>N</u>	SSWS Valves to Turbine Generator Area	<u>N</u>	Any Diesel Generator
<u>N</u>	RHR Suction Line valves from Hot Leg	<u>N</u>	Auxiliary Feed Pump
<u>N</u>	CVCS Letdown line Control and Isolation Valves	<u>N</u>	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

<u>Y/N</u>	<u>OPERATOR ACTION</u>
<u>N</u>	Restore AC power during SBO
<u>N</u>	Connect to gas turbine
<u>N</u>	Trip Reactor and RCPs after loss of component cooling system
<u>Y</u>	Re-align RHR system for re-circulation
<u>N</u>	Un-isolate the available CCW Heat Exchanger
<u>N</u>	Isolate the CVCS letdown path and transfer charging suction to RWST
<u>N</u>	Cooldown the RCS and depressurize the system
<u>N</u>	Isolate the affected Steam Generator that has the tube rupture(s)
<u>N</u>	Early depressurize the RCS
<u>N</u>	Initiate feed and bleed

Complete this evaluation form for each ESG.